

Francium

Fr (Fr I). Strongest classified transitions below 2 μm (NIST): 52 levels, 70 lines.

Fr+ ionisation limit 32848.87 cm⁻¹ = 4.07274 eV (304.424 nm)



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
7s ² S _{1/2} – 8p ² P _{3/2} ^o	422.6845	422.5655	709.2582	0.648	2.820e+06	-	-	
7s ² S _{1/2} – 8p ² P _{1/2} ^o	432.6577	432.5361	692.9091	0.459	2.640e+06	-	-	
7p ² P _{1/2} ^o – 10d ² D _{3/2}	553.3253	553.1716	541.8015	0.945	2.670e+06	-	-	
7p ² P _{3/2} ^o – 13d ² D _{5/2}	563.9154	563.7589	531.6267	1.117	2.350e+06	-	-	
7p ² P _{1/2} ^o – 9d ² D _{3/2}	585.5114	585.3491	512.0182	1.326	4.440e+06	-	-	
7p ² P _{3/2} ^o – 10d ² D _{5/2}	609.6964	609.5276	491.7078	1.546	3.560e+06	-	-	
7p ² P _{3/2} ^o – 9d ² D _{5/2}	648.6002	648.4210	462.2146	1.777	3.910e+06	-	-	
7p ² P _{1/2} ^o – 8d ² D _{3/2}	650.9040	650.7242	460.5786	2.092	8.040e+06	-	-	
7s ² S _{1/2} – 7p ² P _{3/2} ^o	718.1845	717.9866	417.4310	5.912	4.780e+07	-	-	
7p ² P _{3/2} ^o – 8d ² D _{5/2}	728.7899	728.5892	411.3565	2.818	6.930e+06	-	-	
7p ² P _{1/2} ^o – 9s ² S _{1/2}	744.4026	744.1976	402.7289	1.058	2.750e+06	-	-	
7s ² S _{1/2} – 7p ² P _{1/2} ^o	817.1664	816.9418	366.8683	4.165	3.220e+07	-	-	
7p ² P _{1/2} ^o – 7d ² D _{3/2}	832.8738	832.6449	359.9494	4.351	1.660e+07	-	-	
7p ² P _{3/2} ^o – 9s ² S _{1/2}	851.2810	851.0472	352.1663	1.493	3.660e+06	-	-	
7p ² P _{3/2} ^o – 7d ² D _{5/2}	960.7134	960.4499	312.0519	5.820	1.290e+07	-	-	
7p ² P _{3/2} ^o – 7d ² D _{3/2}	968.9892	968.7236	309.3868	1.938	2.090e+06	-	-	

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

Arrow label: vacuum wavelength | reduced dipole matrix element |⟨J||er||J'⟩| in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation
Level caption: NIST energy and measured lifetime (where one exists).